Artificial Intelligence CoSc4011_

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Artificial Intelligence

- AI: study of the mental faculties through the use of computational models
  - concerned with the design of intelligence in an artificial device
  - Intelligence tasks involving higher mental process; the computational part of the ability to achieve goals
    - creativity, solving problems, pattern recognition, classification, classification,
    - learning, induction, deduction, building analogies,
    - optimization language processing, knowledge and many more.

Artificial Intelligence

- AI - Interpretations of Intelligence:
  1. Think human like: AI is about designing systems that are as intelligent as humans; emulate the human thought process; cognitive science approach to AI.
     - Cognitive science approach
  2. Act humanly: 'imitation game', interrogator would not know if it was interrogating an AI or a human (Turing test)
     - Turing test approach

Artificial Intelligence

- AI - Interpretations of Intelligence:
  3. Think rationally: soundness and completeness of the inference mechanism of it's Logic (how the system arrives at a conclusion; reasoning behind its selection of actions)
     - Law of thought approach
  4. Act rationally: AI as the study of rational agents (how the system acts and performs, and not so much about the reasoning process)
     - Rational agent approach

Artificial Intelligence

- AI:
  - Intelligent behavior: tasks and applications
    - Perception: image recognition, computer vision
    - Reasoning
    - Learning
    - Natural language processing, speech processing
    - Solving problems
    - Robotics
    - Thinking
    - Acting - in the complex environment
    - Knowledge – applying successfully in new situation

Artificial Intelligence

- AI - Approaches to AI
  - Strong AI: build self-aware machines that can truly reason and solve problems
    - indistinguishable from that of a human being.
  - Weak AI: cannot truly reason and solve problems, but can act as if it were intelligent.
    - suitably programmed machines can simulate human cognition.
  - Applied AI: produce commercially viable smart system (i.e. recognize faces)
    - already enjoyed considerable success.
  - Cognitive AI: test theories about how the mind works on computers
    - solve abstract problems.

Artificial Intelligence

- AI – AI Techniques
  - Act intelligent techniques
    - Describe and match
    - Constraint satisfaction
    - Generate and Test
    - Goal reduction
    - Tree search
    - Rule based system
  - Biology inspired
    - Neural Network
    - Reinforcement learning
    - Genetic Algorithms

Artificial Intelligence

- Intelligent Agents
  - AI system: composed of an agent and its environment.

Artificial Intelligence

- Intelligent Agents – AI system
  - Agent: can perceive it’s environment (sensors) and act upon that environment (effectors)
    - Autonomous agent: decides autonomously which action to take in the current situation to maximize progress towards its goals
    - Agent = Architecture + Agent Program
      - Architecture: the machinery that an agent executes on
      - Agent Program: an implementation of an agent function
    - Autonomy: extent of an agent’s behavior is determined by its own experience, rather than knowledge of designer
      - No autonomy: ignores environment/data
      - Complete autonomy: must act randomly/no program
      - Ideal: design agents to have some autonomy
        - Possibly become more autonomous with experience

Artificial Intelligence

- Intelligent Agents – AI system
  - environment may contain other agents.
  - most famous artificial environment: the Turing Test environment
  - Rationality: status of being reasonable, sensible, and having good sense of judgment

Artificial Intelligence

- Intelligent Agents – AI system – Rationality:
  - concerned with expected actions and results depending upon what the agent has perceived
  - Ideal Rational Agent: agent capable of doing expected actions to maximize its performance measure, on the basis of
    - Its percept sequences
    - Its built-in knowledge bases

Artificial Intelligence

- Intelligent Agents - AI system - <u>Rationality</u>:
  - rationality depends on:
    - performance measures, (determine the degree of success)
    - agent's Percept Sequence till now.
    - agent's posterior knowledge about the environment.
    - actions that the agent can carry out.
  - A rational agent always performs right action (right action is which makes agent most successful in the given percept sequence)

Artificial Intelligence

- Intelligent Agents – AI system:
  - Omniscience: state of possessing unlimited knowledge about all things possible
    - knows the actual effects of its actions
    - is impossible in real world

Artificial Intelligence

- Intelligent Agents:
  - Performance Measure of Agent: criteria determining how successful an agent is
    - subjective measure
    - in terms of:
      - speed or efficiency
      - accuracy or the quality of the solutions achieved
      - power usage, money, …
  - Behavior of Agent: action performed after any given sequence of percepts.

Artificial Intelligence

- Intelligent Agents:
  - Percept: perceptual inputs (complete set of inputs) at a given instance;
  - Percept Sequence: history of all that the agent has perceived till date.
  - Agent Function: map from the precept sequence to an action.
  - Action: operation involving an effector

Artificial Intelligence

- Intelligent Agents – Properties of Task environments:
  - Discrete / Continuous: whether there are a limited number of distinct, clearly defined, states of the environment or not
    - i.e. chess - discrete; continuous - driving
  - Observable / Partially Observable: whether it is possible to determine the complete state of the environment at each time point from the percepts (observable) or not (partially observable)
  - Static / Dynamic: whether environment changes while an agent is acting(dynamic) or not (static)

Artificial Intelligence

- Intelligent Agents – Properties of Task environments:
  - Single agent / Multiple agents
  - Accessible / Inaccessible – whether the agent's sensory apparatus can have access to the complete state of the environment or not.
  - Deterministic / Non-deterministic – If the next state of the environment is completely determined by the current state and the actions of the agent, then the environment is deterministic; otherwise it is non-deterministic.

Artificial Intelligence

- Intelligent Agents - Properties of Task environments:
  - Episodic / Non-episodic
    - episodic environment: each episode consists of the agent perceiving and then acting.
      - quality of its action depends just on the episode itself.
      - subsequent episodes do not depend on the actions in the previous episodes.
      - much simpler because the agent does not need to think ahead

Artificial Intelligence

- Intelligent Agents – Agent Types
  - Simple reflex agent: simplest agents,
    - decisions on the basis of the current percept and ignore the rest of the percept history
    - these agents only succeed in the fully observable environment
    - works on Condition-action rule: maps the current state to action
    - very limited intelligence, not adaptive to changes in the environment
    - i.e. Room Cleaner agent

Artificial Intelligence

- Intelligent Agents – Agent Types
  - Model-based reflex agent
    - can work in a partially observable environment, and track the situation
    - factors:
      - Model: knowledge about "how things happen in the world,"
      - Internal State: a representation of the current state based on percept history
    - updating the agent state requires information about:
      - how the world evolves
      - how the agent's action affects the world

Artificial Intelligence

- Intelligent Agents – Agent Types
  - Goal-based agent: agent needs to know its goal which describes desirable situations
    - expand the capabilities of the model-based agent by having the "goal”
    - may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not (searching and planning)

Artificial Intelligence

- Intelligent Agents – Agent Types
  - Utility-based agent
    - similar to the goal-based agent but provides utility measurement
    - utility measurement provides a measure of success at a given state
    - agent acts based not only goals but also the best way to achieve the goal
    - useful when there are multiple possible alternatives and has to choose best action
    - utility function maps each state to a real number (efficiency)

Artificial Intelligence

- Intelligent Agents – Agent Types
  - Learning agent
    - learning allows an agent to operate in initially unknown environments, required for true autonomy
    - agent can learn from its past experiences; starts to act with basic knowledge, then it’s able to act and adapt automatically
    - learning element modifies the performance element

Artificial Intelligence

- Intelligent Agents – Agent Types
  - Learning agent – Conceptual components:
    - Learning element: for making improvements by learning from environment
    - Critic Learning element: takes feedback from critic (how well the agent is doing with respect to a fixed performance standard)
    - Performance element: for selecting external action
    - Problem generator: for suggesting actions that will lead to new and informative experiences.

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching: step by step procedure to solve a search-problem in a given search space.

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching - Factors:
    - Search Space: represents a set of possible solutions, which a system may have.
    - Start State: a state from where agent begins the search.
    - Goal test: function observing current state & if the goal state is achieved or not.

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching:
    - Search tree: representation of search problem; root corresponds to the initial state.
    - Actions: description of all the available actions to the agent.
    - Transition model: description of what each action d
    - Path Cost: It is a function which assigns a numeric cost to each path. Solution: It is an action sequence which leads from the start node to the goal node. Optimal Solution: If a solution has the lowest cost among all solutions.

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching - Strategies:
    - Uninformed Search Strategies (<u>level order traversal</u>): does not contain any domain knowledge such as closeness and the location of the goal
      - operates in a brute-force way
      - applies a way in which search tree is searched without any information about the search space (blind search)
      - examines each node of the tree until it achieves the goal node

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Uninformed Search:
    - DFS
      - requires very less memory
      - may go to the infinite loop
      - Complete within finite state space, non-optimal
      - Time Complexity - $O(n^m)$: m = maximum depth; n – branching factor; in other words: $O(b^d)$ (exponential)
      - Space complexity - O(n) (linear)

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching - Strategies - Uninformed Search:
    - BFS
      - will provide a solution if any solution exists
      - provides the minimal solution which requires the least number of steps if there is more than one solution
      - requires lots of memory since each level of the tree must be saved into memory
      - needs lots of time if the solution is far away from the root node
      - Complete since BFS will find the solution, optimal
      - b - branching factor - **maximum number of successors (children)** any node can have (2 for binary tree)
      - Time Complexity - $O(b^d)$: d = depth of shallowest solution, b = node at every state; Space complexity - $O(b^d)$ – exponential

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Uninformed Search:
    - Depth limited search
      - similar to DFS with a predetermined limit
      - Complete if the solution is above the depth-limit (else incomplete), may not be optimal for more than one solution
      - Time complexity – $O(b^l)$ – exponential
      - Space complexity – $O(b * l)$ – linear

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Uninformed Search:
    - Uniform cost search
      - for traversing a weighted tree or graph
      - find a path to the goal node which has the lowest cumulative cost
      - equivalent to BFS algorithm if the path cost of all edges is the same
      - Complete, optimal because at every state the path with the least cost is chosen

Artificial Intelligence

- Intelligent Agents - Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Uninformed Search:
    - Uniform cost search - Time complexity:
      - b - branching factor - **maximum number of successors (children)** any node can have (2 for binary tree)
      - **C*** is the cost of the optimal solution (i.e. least cost to reach goal = 10)
      - **ε** is the minimum positive cost of any action (i.e. smallest costing edge = 2)
      - **maximum number of steps** (i.e. edge traversals) to reach a cost of C* : $C^* / \boldsymbol{\varepsilon}$
      - node level: $[C^* / \boldsymbol{\varepsilon}]$ + 1 including root node
      - $O\left(b^0 + b^1 + b^2 + \ldots + b^{[C^* / \varepsilon]}\right) = O(b^{1+[C^* / \varepsilon]})$ - <u>exponential</u>
        - for initial node - $b^0 = 1$ node
        - for second level - $b^1 = b = 2$ for binary tree (2 nodes)
        - for third level - $b^2 = 2^2 = 4$ for binary tree (4 nodes) … until maximum depth traversal $C^* / \boldsymbol{\varepsilon}$

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Uninformed Search:
    - Uniform cost search:
      - Space complexity: same as Time complexity - exponential

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Uninformed Search:
    - Iterative Deeping DFS
      - combination of DFS and BFS algorithms
      - finds out the best depth limit and does it by gradually increasing the limit until a goal is found
      - combines the benefits of BFS and DFS search algorithm in terms of fast search and memory efficiency
      - main drawback of IDDFS is that it repeats all the work of the previous phase
      - Complete if branching factor is finite, optimal if path cost is non-decreasing from depth
      - Time Complexity - $O(b^d)$: d = depth of shallowest solution, b = node at every state (branching factor); <u>exponential</u>
      - Space complexity - $O(b * d)$ - <u>linear</u>

Artificial Intelligence

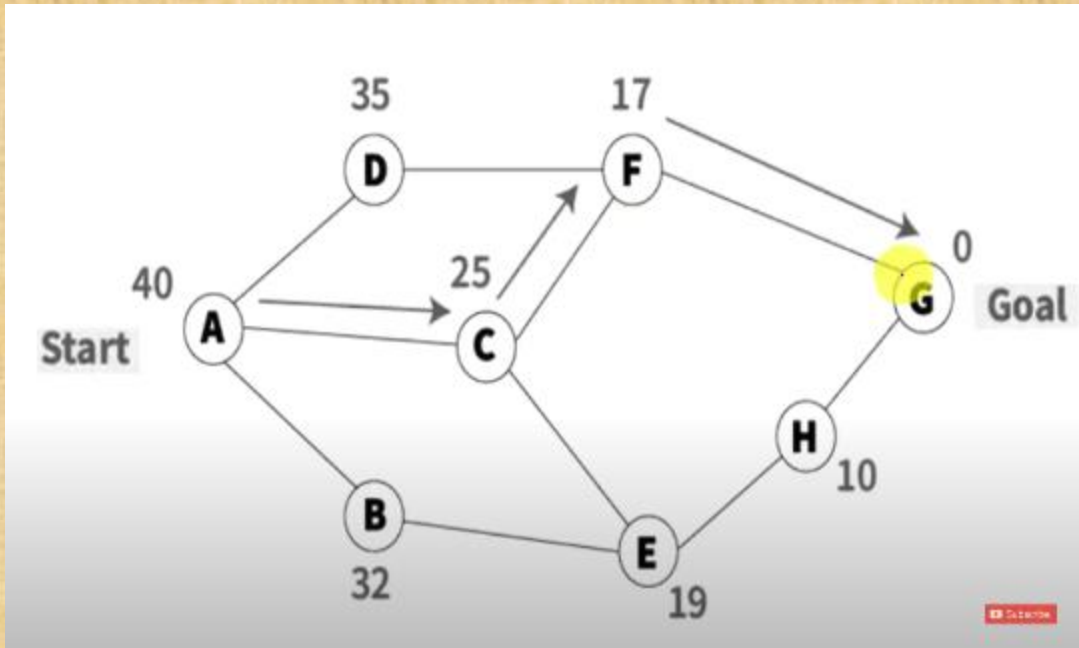
- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Uninformed Search:
    - Bidirectional search
      - runs two simultaneous searches, one starts the search from an initial vertex and other starts from goal vertex
      - search stops when these two graphs intersect each other
      - can use search techniques such as BFS, DFS, DLS, etc…
      - fast, requires less memory
      - implementation is difficult, goal state should be known in advance
      - Same complexity as the used technique, i.e. same as BFS if BFS was used bi-directionally

Artificial Intelligence

- Intelligent Agents - Problem Solving (Goal Based) Agent - Problem Solving by Searching:
  - Searching - Strategies - Informed Search - Best first search
    - Greedy Best first search: always selects the path which appears best at that moment
      - https://www.youtube.com/watch?v=GVvN0ikNekw
      - combination of BFS and DFS
      - at each step, choose the most promising node (expand the node which is closest to the goal node estimated by heuristic function)
        - h(n)= estimated cost from node n to the goal
      - more efficient than BFS and DFS algorithms
      - can behave as an unguided DFS in the worst-case scenario, and can get stuck in a loop as DFS
      - incomplete even if the given state space is finite, not optimal
      - same complexity as BFS in worst case - exponential

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – Best first search
    - Greedy Best first search: always selects the path which appears best at that moment

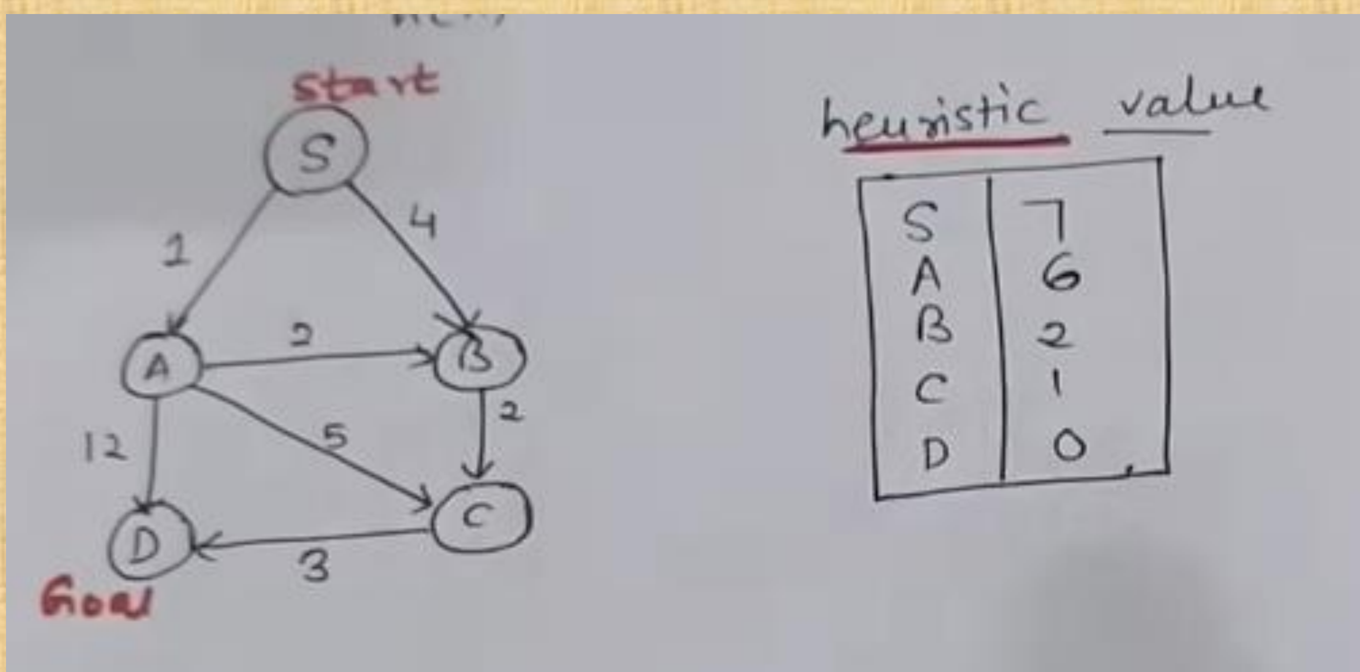


Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Informed Search - Best first search
    - A * search / A star search: most commonly known form of best-first search
      - https://www.youtube.com/watch?v=Eiu1Cb-veCc
      - combined features of UCS and greedy best-first search
        - similar to UCS except instead of g(n), it uses g(n)+h(n) = fitness number;
      - best algorithm than other search algorithms
      - does not always produce the shortest path as it mostly based on heuristics
      - complete, optimal
      - same complexity as BFS

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – Best first search
    - A * search / A star search



Artificial Intelligence

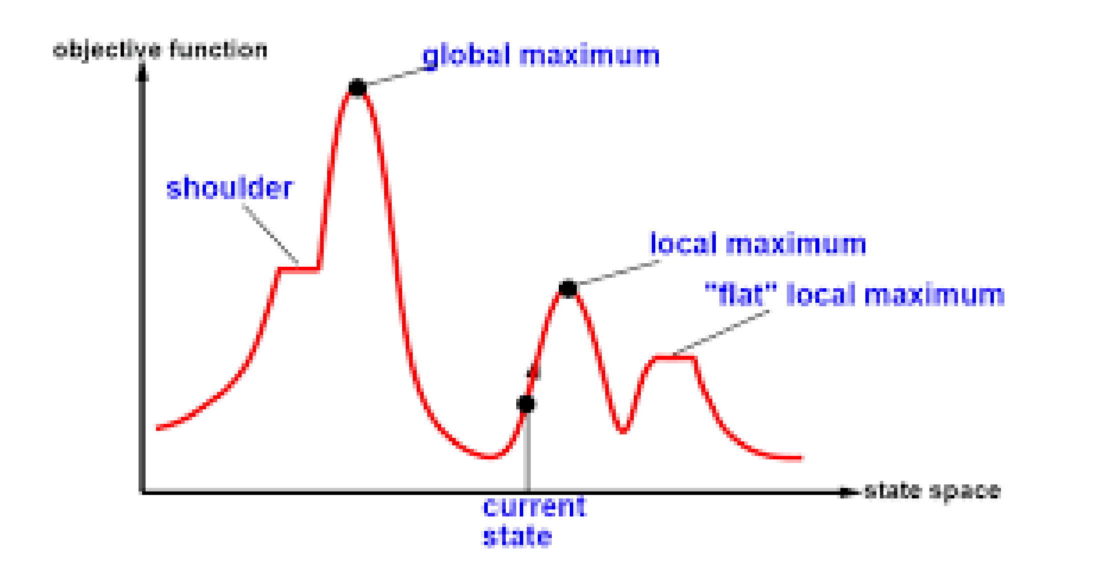
- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Informed Search
    - Local Search Strategies: path to the goal is irrelevant; the goal state itself is the solution; widely used for very big problems; good but not optimal solutions; very memory efficient (1 or a few states)

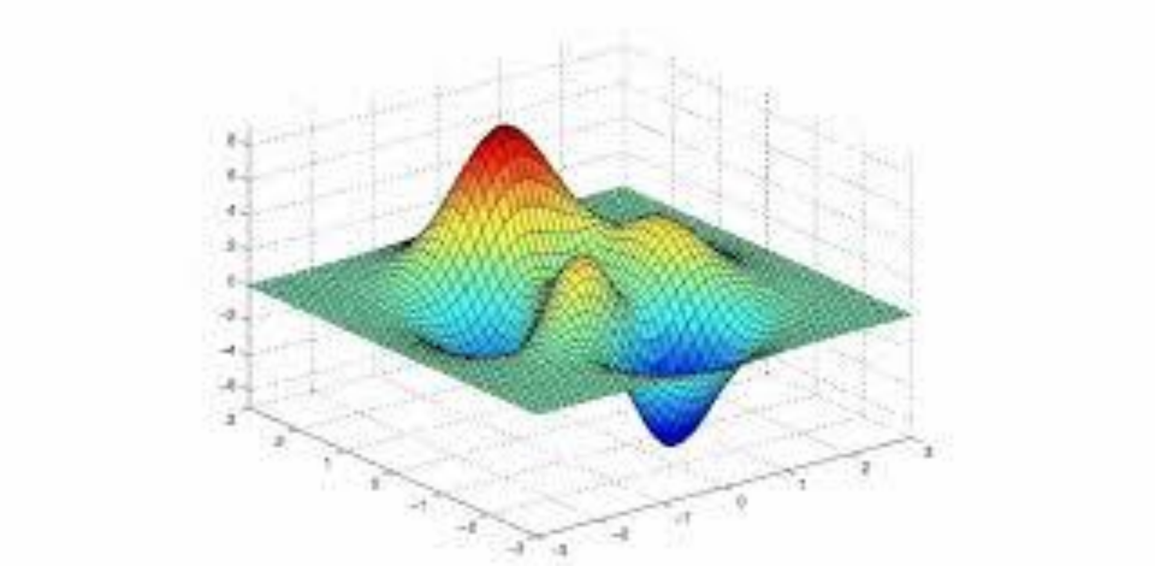
Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – Local Search Strategies:
    - Hill climbing algorithm (greedy local search):
      - moves in the direction of increasing value to find the peak (best solution);
      - terminates where no neighbor has a higher value;
      - used when a good heuristic is available;
      - only keeps a single current state (no need of search tree)
      - no backtracking and doesn't remember previous states
      - shoulder - It is a plateau region which has an uphill edge

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – Local Search Strategies:
    - Hill climbing algorithm (greedy local search):
      - state-space landscape: graph between various states and cost



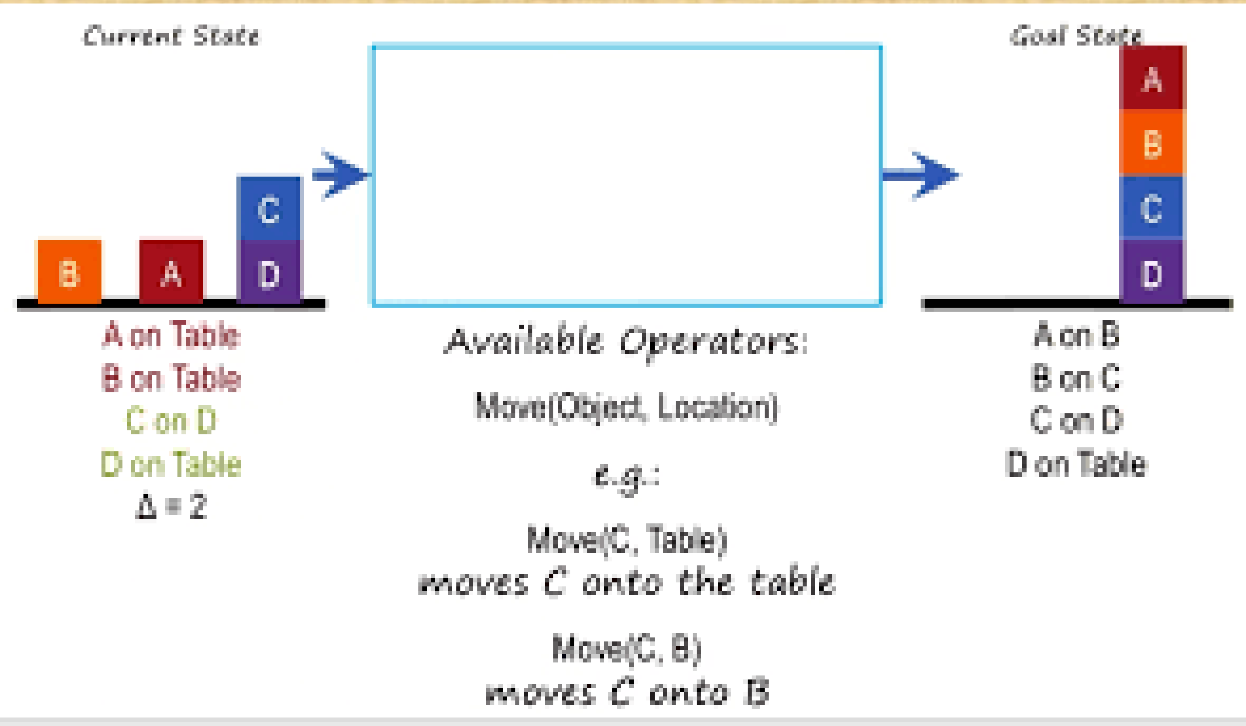


Artificial Intelligence

- Intelligent Agents - Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies - Informed Search - Local Search Strategies:
    - Hill climbing algorithm (greedy local search):
      - Types:
        1. Simple hill climbing algorithm: only evaluates the neighbor node state at a time; selects the first one which optimizes current cost
        2. Steepest-Ascent algorithm: examines all the neighboring nodes of the current state; select neighbor closest to the goal
        3. Stochastic hill climbing: selects one node at random and decides whether it should be expanded

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – Local Search Strategies:
    - Means ends analysis: first solve the major part of a problem and then go back and solve the small problems
      - evaluates the difference between the current state and goal state



Artificial Intelligence

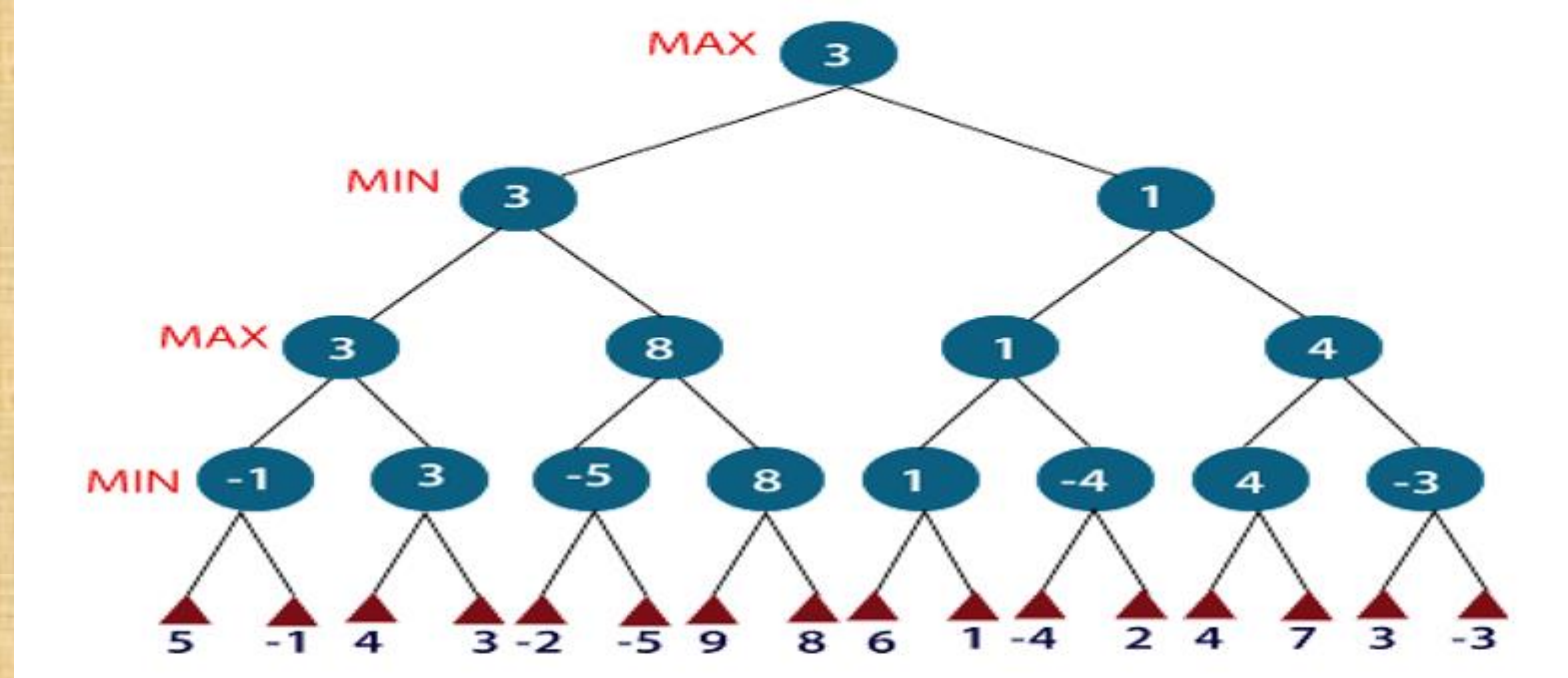
- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search
    - Adversarial Search Strategies: examine problem when we try to plan ahead of the world and other agents are planning against us
      - https://www.youtube.com/watch?v=l-hh51ncgDI
      - two or more players with conflicting goals are trying to explore the same search space for the solution

Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – Adversarial Search Strategies
    - Minimax algorithm: recursive or backtracking algorithm; used in decision making and game theory; DFS algorithm
      - two players play the game, one is MAX and other is MIN
      - MIN: Decrease the chances of MAX to win the game
      - MAX: Increases his chances of winning the game
      - game is played alternatively between them

Artificial Intelligence

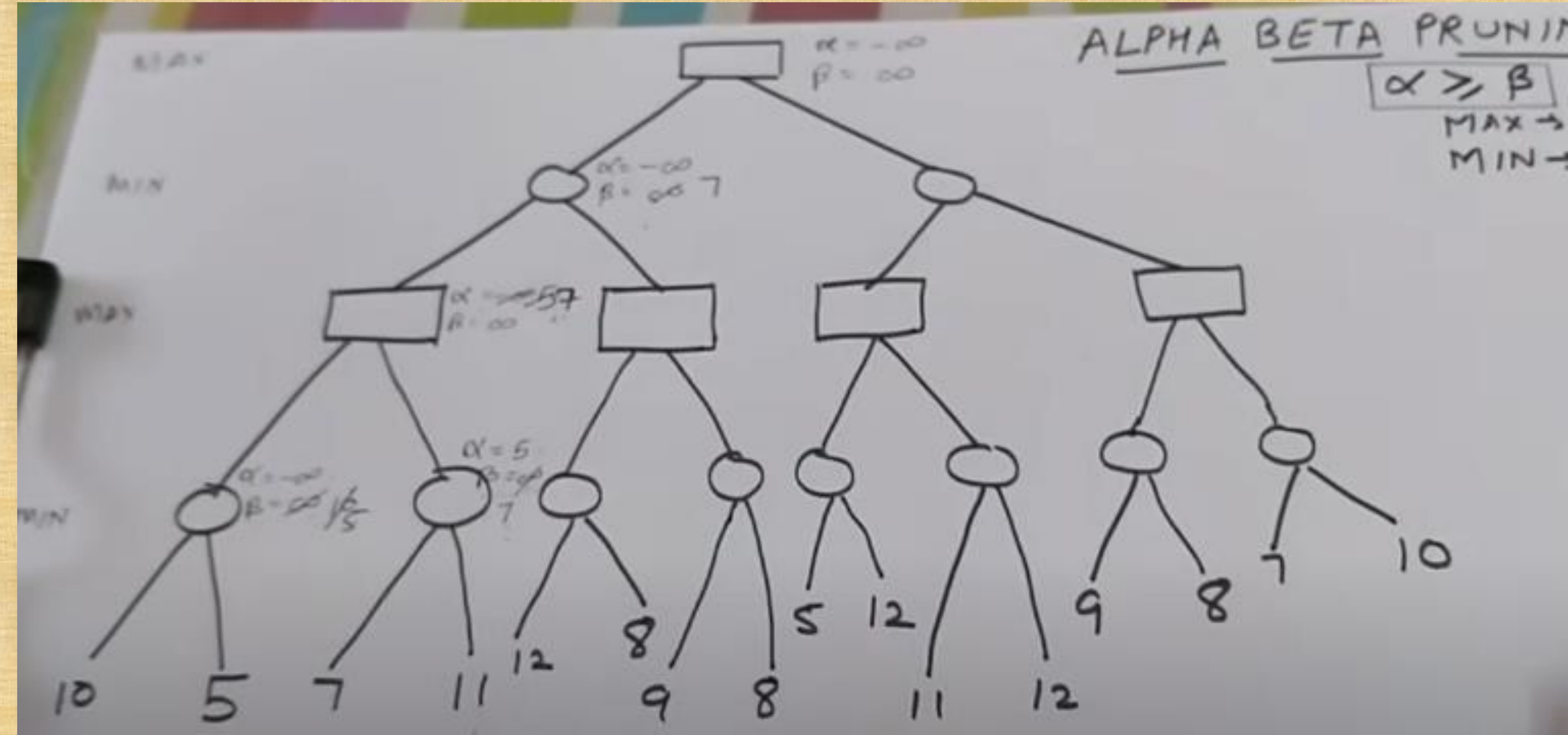
- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – <u>Adversarial Search Strategies</u>
    - Minimax algorithm



Artificial Intelligence

- Intelligent Agents – Problem Solving (Goal Based) Agent – Problem Solving by Searching:
  - Searching – Strategies – Informed Search – Adversarial Search Strategies
    - Alpha Beta Pruning
      - https://www.youtube.com/watch?v=_i-lZcbWkps
      - modified version of the minimax algorithm
      - Alpha: highest-value choice we have found so far at any point along the path of Maximizer (initial value = -∞)
      - Beta: lowest-value we have found so far at any point along the path of Minimizer (initial value = +∞)
      - Max player will only update the value of alpha; Min player will only update the value of beta
      - main condition for alpha-beta pruning: $\alpha >= \beta$

Artificial Intelligence



Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - agents who have the capability of maintaining an internal state of knowledge, reason over that knowledge, update their knowledge after observations and take actions
  - intelligent agent needs knowledge about the real world for <u>taking decisions</u> and <u>reasoning</u> to act efficiently
  - Logical agent: agents with some representation of complex knowledge about the world or its environment
    - uses the inferences to derive new information from the knowledge combined with new inputs

Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - Knowledge: a sentence in a knowledge representation language (formal language)
  - Levels of knowledge
    1. Knowledge level (1st level)
       - specify what the agent knows, and what the agent goals are
    2. Logical level
       - sentences are encoded into different logics
    3. Implementation level
       - agent performs actions as per logical and knowledge level

Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - Knowledge Base: set of sentences in a formal language representing facts about the world.
    - central component of a knowledge-based agent
    - collection of sentences
  - Inference: deriving new sentences from old.
    - Inference system: allows us to add a new sentence to the knowledge base
    - Inference system rules:
      - Forward chaining (facts to goals, data driven)
      - Backward chaining (goals to facts, goal driven)

Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - KB agent must able to do
    - represent states, actions...
    - incorporate new percepts.
    - update the internal representation of the world.
    - deduce the internal representation of the world,
    - deduce appropriate actions.
  - Approaches to design KB agents
    - Declarative approach: feed necessary information in an empty knowledge-based system
    - Procedural approach: knowledge stored in empty system in the form of program code
    - a successful agent can be built by combining both

Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - Knowledge Representation
    - Object: All the facts about objects in our world domain.
    - Events: the actions which occur in our world.
    - Performance: behavior which involves knowledge about how to do things.
    - Meta-knowledge: knowledge about what we know.
    - Facts: truths about the real world and what we represent.

Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - Techniques of Knowledge Representation
    - Logical Representation: drawing a conclusion based on various conditions
      - Logic Programming: language with some concrete rules which deals with propositions
      - consists of precisely defined syntax and semantics
        - Syntax: constructing legal sentences
        - Semantics: rules to interpret the sentence

Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - Techniques of Knowledge Representation
    - Frame Representation
      - Frame: a record/row for an object
      - Slot: attribute (column)
      - Facet: metadata about the slot (i.e. type)
      - makes the programming easier by grouping the related data
      - comparably flexible and used by many applications in AI
      - inference mechanism is not easy to process

Artificial Intelligence

- Knowledge Representation and Reasoning - Knowledge based agent
  - Techniques of Knowledge Representation
    - Production Rules
      - checks for the condition and if the condition exists then production rule fires and corresponding action is carried out
      - production rules are highly modular, so we can easily remove, add or modify an individual rule
      - does not exhibit any learning capabilities

Artificial Intelligence

- Propositional Logic: the simplest form of logic where all the statements are made by propositions
  - proposition: a declarative statement which is either true or false
    - tautology: always true
    - contradiction: always false
    - contingency: has both true and false values
    - types:
      - Atomic Proposition: a single proposition symbol.
      - Compound proposition: constructed by combining simpler or atomic propositions, using parenthesis and logical connectives
  - PL consists of an object, relations or function, and logical connectives(operators)
  - Statements which are questions, commands, or opinions are not propositions

Artificial Intelligence

- Propositional Logic:
  - Logical connectives (mainly five):
    - Negation: $\neg$ P
    - Conjunction: P $\land$ Q
    - Disjunction: P $\lor$ Q
    - Implication (if-then rules): P $\rightarrow$ Q
    - Bi-conditional: P$\Leftrightarrow$ Q
  - Precedence of connectives:
    - parenthesis $\rightarrow$ negation $\rightarrow$ conjunction $\rightarrow$ dis-junction $\rightarrow$ Implication $\rightarrow$ bi-implication

Artificial Intelligence

- Propositional Logic:
  - Inference in Propositional Logic:
    - generating the conclusions from evidence and facts
    - Inference rules: the templates for generating valid arguments; applied to derive proofs
    - Proof: a sequence of the conclusion that leads to the desired goal

Artificial Intelligence

- Propositional Logic:
  - Inference in Propositional Logic:
    1. Modus Ponens: if P and $P \rightarrow Q$ is true, then we can infer that Q will be true
    2. Modus Tollens: if $P \rightarrow Q$ is true and $\neg Q$ is true, then $\neg P$ will also true
    3. Hypothetical Syllogism: if $P \rightarrow Q$ is true and $Q \rightarrow R$ is true, then $P \rightarrow R$ is true
    4. Disjunctive Syllogism: if $P \vee Q$ is true, and $\neg P$ is true, then Q will be true
    5. Addition: if P is true, then $P \vee Q$ will be true.
    6. Simplification: if $P \wedge Q$ is true, then Q and P will also be true
    7. Resolution: if $P \vee Q$ and $\neg P \wedge R$ is true, then $Q \vee R$ will also be true

Artificial Intelligence

- Predicate (First-Order) Logic
  - Inference in First-Order Logic: used to deduce new facts or sentences from existing sentences
  - Terminologies
    - Substitution: performed on terms and formulas; occurs in all inference systems in FOL.
      - complex in the presence of quantifiers in FOL.
      - F[a/x], so it refers to substitute a constant "a" in place of variable "x”
    - Equality-FOL: also uses equality (two terms refer to the same object) in addition to predicates and terms.
      - Brother (John) = Smith

Artificial Intelligence

- Predicate (First-Order) Logic
  - Terminologies
    - Universal Generalization: valid inference rule, states that if premise P(c) is true for any arbitrary element c in the universe of discourse, then $\forall$ x P(x) is true (for all elements)
    - Universal Instantiation: state that we can infer any sentence P(c) by substituting a ground term c (a constant within domain x) from $\forall$ x P(x) for any object in the universe of discourse.
      - can be applied multiple times to add new sentences; new KB is logically equivalent to the previous KB.
      - IF "Every person like ice-cream"=> $\forall$x P(x) so we can infer that: "John likes ice-cream" => P(c)

Artificial Intelligence

- Predicate (First-Order) Logic
  - Terminologies
    - Existential introduction(existential generalization): If there is some elements C in the universe of discourse which has a property P, then we can infer that there exists something in the universe which has the property P
      - generalizing from a specific fact to a broad existential claim
      - Priyanka got good marks in English. Therefore, someone got good marks in English.

Artificial Intelligence

- Predicate (First-Order) Logic
  - Terminologies
    - Existential Instantiation (Existential Elimination): can be applied only once to replace the existential sentence; the new KB is not logically equivalent to old KB, but it will be satisfiable if old KB was satisfiable.
      - states that one can infer P(c) from the formula given in the form of ∃x P(x) for a new constant symbol c - "There exists an x such that P(x) is true. Let's pick a specific (new) constant c that has property P"
      - i.e. ∃x Crown(x) ∧ OnHead(x, John): There exists something that is a crown and is on John's head.
      - you don't know **what** that crown is, but you know **something** exists, so introduce a new name, say **K**, and write: Crown(K) ∧ OnHead(K, John) - <u>Skolemization</u> process.
        - K - <u>Skolem</u> constant

Artificial Intelligence

- Predicate (First-Order) Logic
  - Unification in FOL: process of making two different logical atomic expressions identical by finding a substitution.
    - The UNIFY algorithm is used for unification
    - i.e. two different expressions: P(x, y), P(a, f(z))
      - substitute x with a, and y with f(z) in the first expression, and it will be represented as a/x and f(z)/y
      - substitution set will be: [a/x, f(z)/y]

Artificial Intelligence

- Predicate (First-Order) Logic
  - Unification in FOL - Conditions for Unification:
    - Predicate symbol must be same, atoms or expression with different predicate symbol can never be unified. Likes(Alice, x) and Loves(Alice, x) Cannot be unified (Likes ≠ Loves)
    - Number of Arguments in both expressions must be identical.
    - It will fail if there are two similar variables present in the same expression.
      - P(x) and P(f(x)) would require x = f(x), which creates an infinite loop

Artificial Intelligence

- Resolution in FOL
  - Resolution: theorem proving technique; builds refutation proofs, i.e., proofs by contradictions.
    - used, if various statements are given, and need to prove a conclusion of those statements.
    - Unification is a key concept in proofs by resolutions.
    - can efficiently operate on the conjunctive normal form or clausal form.
    - Clause (a unit clause): Disjunction of literals (an atomic sentence).
    - Conjunctive Normal Form: conjunction of clauses.

Artificial Intelligence

- Resolution in FOL
  - Steps for Resolution:
    1. Conversion of facts into first-order logic.
    2. Convert FOL statements into CNF
    3. Negate the statement which needs to prove (proof by contradiction)
    4. Draw resolution graph (unification)

Artificial Intelligence

- Reasoning: mental process of deriving logical conclusion and making predictions from available knowledge, facts, and beliefs; it is
  - a way to infer facts from existing data.
  - general process of thinking rationally, to find valid conclusions.

Artificial Intelligence

- Reasoning:
    1. Deductive reasoning: deducing new information from logically related known information.
        - the argument's conclusion must be true when the premises are true
        - top-down reasoning, and contradictory to inductive reasoning
        - mostly starts from the general premises to the specific conclusion

Artificial Intelligence

- Reasoning:
  2. Inductive reasoning: arrive at a conclusion using limited sets of facts by generalization.
     - starts with the series of specific facts or data and reaches to a general statement or conclusion.
     - also known as cause-effect reasoning or bottom-up reasoning

Artificial Intelligence

- Reasoning:
  3. Abductive reasoning: starts with single or multiple observations to find the most likely explanation or conclusion for the observation.
     - extension of deductive reasoning, but in abductive reasoning, the premises do not guarantee the conclusion.
     - example
       - Implication: Cricket ground is wet if it is raining
       - Axiom: Cricket ground is wet.
       - Conclusion It is raining

Artificial Intelligence

- Reasoning:
    4. Common Sense Reasoning: informal form of reasoning gained through experiences
        - relies on good judgment rather than exact logic and operates on heuristic knowledge and heuristic rules
        - i.e. If I put my hand in a fire, then it will burn

Artificial Intelligence

- Reasoning:
  5. Monotonic Reasoning: once the conclusion is taken, then it will remain the same even if we add some other information to existing information in our knowledge base
     - not useful for the real-time systems, as in real time, facts get changed, so we cannot use monotonic reasoning
     - used in conventional reasoning systems, and a logic-based system is monotonic
     - any theorem proving is an example of monotonic reasoning

Artificial Intelligence

- Reasoning:
   6. Non-monotonic Reasoning: some conclusions may be invalidated if we add some more information to KB.
      - deals with incomplete and uncertain models
      - example:
         - Birds can fly
         - Penguins cannot fly
         - Pitty is a bird
         - from the above we can conclude that Pitty can fly
         - if we add one another sentence into knowledge base "Pitty is a penguin", which concludes "Pitty cannot fly", it invalidates the above conclusion.

Artificial Intelligence

- Reasoning under uncertainty
  - Uncertainty: unsure about predicates
    - $A \rightarrow B$, means if A is true then B is true, but say we don't know that A is true or not so we cannot express this statement, this situation is called uncertainty
    - Causes of uncertainty
      - Information occurred from unreliable sources.
      - Experimental Errors
      - Equipment fault
      - Temperature variation
      - Climate change.

Artificial Intelligence

- Reasoning under uncertainty
  - Probabilistic reasoning: a way of knowledge representation applying the concept of probability to indicate the uncertainty in knowledge; combine probability theory with logic.
    - Ways to solve problems with uncertain knowledge:
      - Bayes' rule: updates the probability of a hypothesis based on new evidence (posterior)
      - Bayesian Statistics: full inference using Bayes' Rule

Artificial Intelligence

- Reasoning under uncertainty
  - Probabilistic reasoning – Baye's Theorem:
    - example: calculate the event A when event B has already occurred (the probability of A under the conditions of B)
      - when A is given:
        - P(B/A) = P(A^B)/P(A)
        - conditional probability = joint probability / marginal probability
      - when B is given:
        - P(A/B) = P(A^B)/P(B)
      - calculate the value of P(B|A) with the knowledge of P(A|B)
        - P(A^B)= P(A/B) * P(B)
        - P(A^B) = P(B/A) * P(A)
        - P(A/B) = P(A^B)/P(B) = P(B/A) * P(A) / P(B)
        - posterior = likelihood * prior probability / marginal probability

Artificial Intelligence

- Expert System: computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise
  - Capabilities: Advising, Instructing and assisting human in decision making, Demonstrating, Deriving a solution, Diagnosing and Explaining, Interpreting input, Predicting results, Justifying the conclusion, Suggesting alternative options to a problem
  - Not capable of: Substituting human decision makers, Producing accurate output for inadequate knowledge base, Refining their own knowledge

Artificial Intelligence

- Expert System: computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise
  - Applications: Design Domain, Medical Domain (deduce cause of disease), Monitoring Systems, Process Control Systems, Knowledge Domain (finding faults), Finance/Commerce (detect fraud)

Artificial Intelligence

- Expert System:
  - Technologies:
    - Expert System Development Environment
    - Tools: editors and debugging tools; have Inbuilt definitions of model, knowledge representation, and inference design
    - Shells: Java Expert System Shell (JESS), Vidwan, Mumbai
  - Benefits: Availability, Less Production Cost, Less Error Rate, Speed, Steady response
  - Limitations: large systems require high cost and development time; difficult to maintain

Artificial Intelligence

- Expert System:
  - Components - <u>Knowledge bases</u>:
    - contains domain-specific and high-quality knowledge.
    - Components:
      - Factual Knowledge: widely accepted by Engineers in the domain,
      - Heuristic Knowledge: about practice, accurate judgement, evaluation, and guessing.
    - Knowledge Acquisition: formed by readings from various experts, scholars, and the Knowledge Engineers

Artificial Intelligence

- Expert System:
  - Components - Inference Engine:
    - In knowledge-based ES: the Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.
    - In rule-based ES, it applies rules repeatedly to the facts obtained from earlier rule application and adds new knowledge into the knowledge base if required.

Artificial Intelligence

- Expert System:
  - Components - Inference Engine - Methods:
    - Forward Chaining (data-driven): deduces outcome; "What can happen next?"
      - down-up approach; making a conclusion based on known facts or data;
      - starts from initial state and reaches the goals state
    - Backward Chaining (goal-driven): "Why this happened?"
      - uses DFS, goal is broken into sub-goals to prove the facts true.
      - based on modus ponens inference rule

Artificial Intelligence

- Expert System:
  - Components - User Interface:
    - provides interaction between user and the ES
    - forms of explanations: natural language; verbal narrations; rule numbers

Artificial Intelligence

- Expert System:
  - Development (iterative)
    - Identify Problem Domain: suitable for an expert system? cost-effectiveness of the system?
    - Design the System: ES Technology, identify integration with other systems and databases
    - Develop the Prototype: acquire domain knowledge from expert & represent in IF-ELSE
    - Test and Refine the Prototype: by knowledge engineer and end users
    - Develop and Complete the ES: ensure interaction b/n ES & its environment; document
    - Maintain the System: regular review & update; evolve

Artificial Intelligence

- Learning Agents
  - Learning: changes in a system enabling more efficiency next time.
  - computer learns if its performance at tasks in T, as measured by P, improves with experience E
  - Components: Learning Element, Performance Element, Critic, Problem Generator

Artificial Intelligence

- Learning Agents - <u>Types of learning</u>:
  - Supervised learning: we are given a set of (x, y) pairs by a "teacher." (discover the relation)
    - Classification
    - Regression
  - Unsupervised learning: we are only given the x's (discover outputs by itself)
    - Clustering
    - Association
  - Reinforcement learning: feedback-based (reward for right action, penalty for wrong action)

Artificial Intelligence

- Decision tree algorithm: classification and prediction tool having a tree like structure,
  - each internal node denotes a test on an attribute
  - each branch represents an outcome of the test
  - each leaf node (terminal node) holds a class label
  - entropy: randomness in the information being processed; higher = harder to draw conclusions
  - information gain: info gained about a random variable from observing another random variable
    - difference between the entropy of parent node and weighted average entropy of child nodes

Artificial Intelligence

- Decision tree algorithm:
  - Gini impurity: how often a randomly chosen element from the set would be incorrectly labeled if it was randomly labeled according to the distribution of labels in the subset.
  - Algorithms to build a decision tree:
    - CART (Classification and Regression Trees): Gini impurity as metric.
    - ID3 (Iterative Dichotomiser 3): entropy and information gain as metric.

Artificial Intelligence

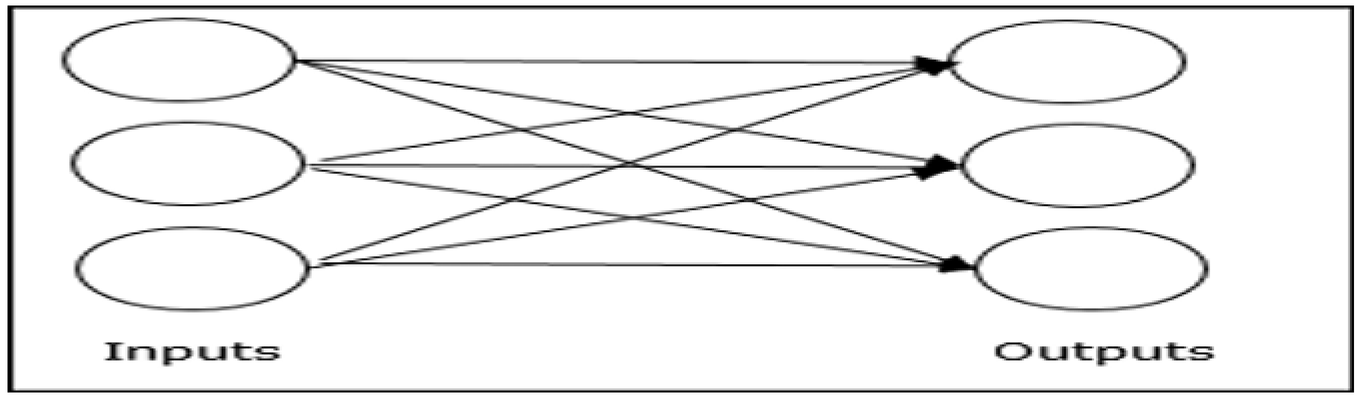
- Regression algorithm: identifying and evaluating the relationship between a dependent variable and one or more independent variables (characteristics measured directly; aka predictor or explanatory variables)

Artificial Intelligence

- **Neural Networks:** approximate a wide range of complex functions representing multi-dimensional input-output maps
  - composed of large number of highly interconnected simple processing elements (neurons)
  - "expert" in the category of information it has been given to analyse
  - used for projections given new situations of interest and answer "what if" questions
  - Advantages: Adaptive learning, Self-Organisation, Real Time Operation (parallel computation), Fault Tolerance via Redundant Information Coding (reduce network without loss of capabilities)

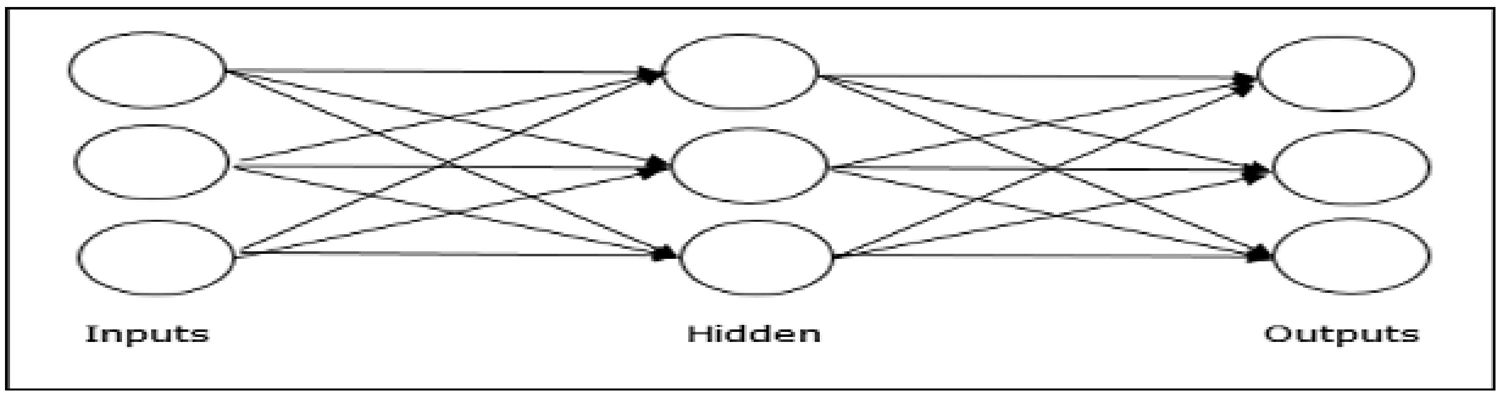
Artificial Intelligence

- **Neural Networks** – Network topology:
  - Feedforward Network: non-recurrent, signal can only flow from input to output
    - Single layer feedforward network: only one weighted layer



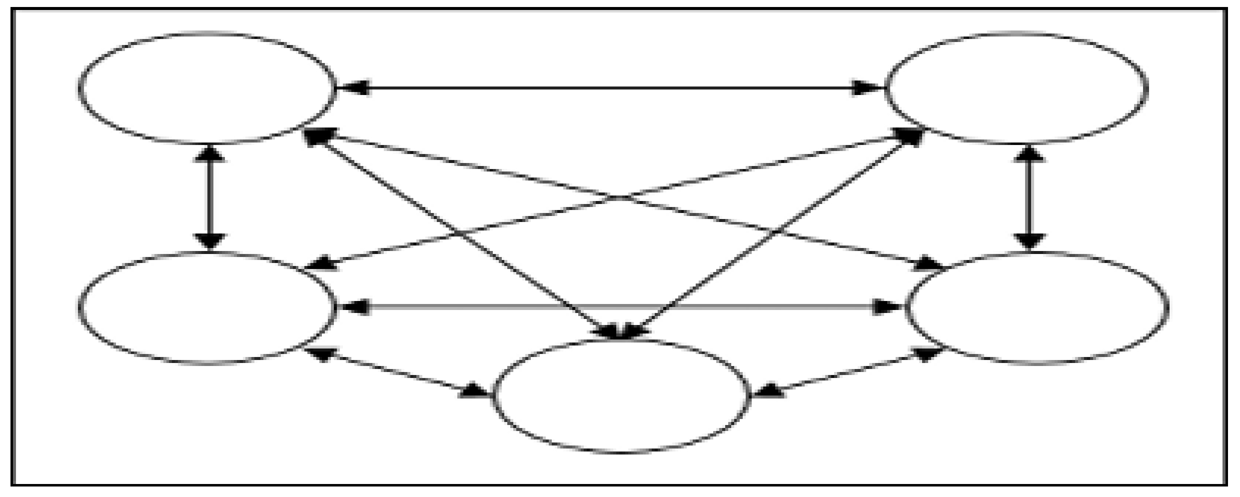
Artificial Intelligence

- **Neural Networks** – Network topology:
  - Feedforward Network:
    - Multilayer feedforward network: one or more hidden layers b/n input and output layer



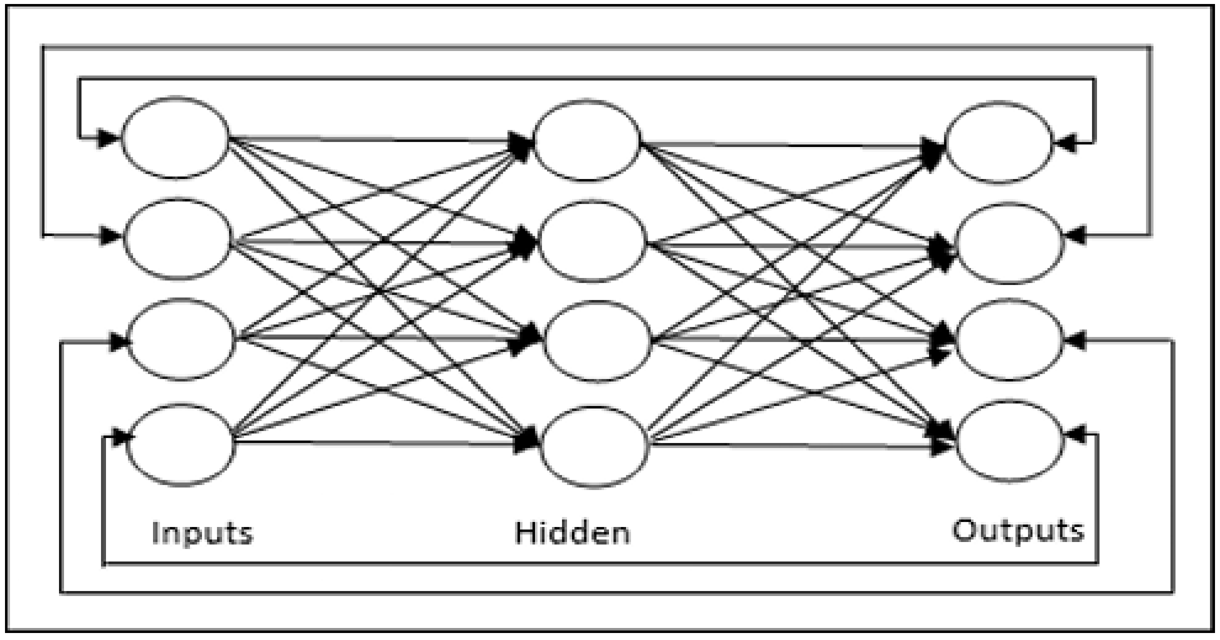
Artificial Intelligence

- **Neural Networks** – Network topology:
  - Feedback network: has feedback paths, non-linear dynamic system
    - Fully recurrent network: all nodes are connected to all other nodes, each node works as both input and output



Artificial Intelligence

- **Neural Networks** – Network topology:
  - Feedback network:
    - Jordan network: closed loop network; output will go to the input again



Artificial Intelligence

- **Neural Networks:**
  - Activation function: the extra force or effort applied over the input to obtain an exact output
    - Linear Activation
    - Sigmoid Activation
      - Binary Sigmoid (Logistic Function) - binary classification, range: (0, 1)
      - Bipolar Sigmoid (Tanh Function) - when negative values are helpful, range: (-1, 1)

Artificial Intelligence

- Communicating, Perceiving, and Acting
  - Natural Language Processing (NLP): used by machines to understand, analyze, manipulate, and interpret human's languages
    - Advantages: users ask about any subject, time efficient, efficient documentation
    - Disadvantages: may not show context, unpredictable

Artificial Intelligence

- Communicating, Perceiving, and Acting – NLP
  - Components:
    - Natural Language Understanding (NLU): extracts metadata (concepts, entities, emotion, relations, semantic rules) for the machine to understand and analyses human language
    - Natural Language Generation (NLG): translator converting the computerized data into natural language representation (generates language)

Artificial Intelligence

- Communicating, Perceiving, and Acting – NLP
  - Phases:
    - Lexical Analysis and Morphological: converts it into meaningful lexemes
    - Syntactic Analysis (Parsing): check grammar, word arrangements and relationships
    - Semantic Analysis: meaning representation
    - Discourse Integration: depends upon the sentences that proceeds it and also invokes the meaning of the sentences that follow it
    - Pragmatic Analysis: discover the intended effect by applying a set of rules that characterize cooperative dialogues

Artificial Intelligence

- Communicating, Perceiving, and Acting – NLP
  - Difficulty of NLP: Ambiguity and Uncertainty exist in the language
    - Lexical Ambiguity: two or more possible meanings within a single word
      - i.e. Manya is looking for a match: match = partner or match = game
    - Syntactic Ambiguity: two or more possible meanings within the sentence
      - i.e. I saw the girl with the binocular; do i have the binocular? or does the girl have it?
    - Referential Ambiguity: when referring to something using a pronoun
      - i.e. Kiran went to Sunita. She said, "I'm hungry."; is it kiran or sunita the hungry one?

Artificial Intelligence

- Communicating, Perceiving, and Acting – NLP
  - Applications: Machine Translation, Spam detection, Sentiment Analysis, Spelling correction, Speech Recognition, Chatbot, Information extraction

Artificial Intelligence

- Robotics: domain in AI that deals with the study of creating intelligent and efficient robots.
  - Robots: artificial agents acting in real world environment.
    - aimed at manipulating the objects by perceiving, picking, moving, modifying the physical properties of object, destroying it, or to have an effect
    - free manpower from repetitive functions without getting bored, distracted, or exhausted

Artificial Intelligence

- Robotics:
  - Locomotion: mechanism that makes a robot capable of moving in its environment

Artificial Intelligence

- Robotics - Locomotion:
  - <u>Legged Locomotion</u>: demonstrates walk, jump, trot, hop, climb up or down
    - consumes more power, more motors, difficult to implement (stability issues)
  - <u>Wheeled Locomotion</u>
    - Standard wheel: Rotates around the wheel axle and around the contact
    - Castor wheel: Rotates around the wheel axle and the offset steering joint
    - Swedish 45o and Swedish 90o wheels: Omni-wheel, rotates around the contact point, around the wheel axle, and around the rollers
    - Ball or spherical wheel: Omnidirectional wheel, difficult to implement
  - <u>Slip/Skid Locomotion</u>: use tracks as in a tank

Artificial Intelligence

- Robotics:
  - Components of robotics:
    - Power Supply, Actuators, Electric motors (rotational movement), Pneumatic Air Muscles (contract 40%), Muscle Wires (contract 5%), Piezo Motors and Ultrasonic Motors (best for industrial robots), Sensors

Artificial Intelligence

- Robotics:
  - Computer vision: technology of AI with which the robots can see; in domains of safety, security, health, access, and entertainment
    - Computer vision tasks:
      - OCR (Optical Character Reader): software converts scanned documents to editable text.
      - Face Detection: to let a user access the software on correct match.
      - Object Recognition
      - Estimating Position: estimate position of object with respect to camera
    - Application of computer vision: Industries, Military, Medicine, Exploration, Entertainment